

Eclipse Environmental Monitoring Summary

Overview

Eclipse is a wireless, solar-powered environmental monitoring device designed for deployment in hyperlocal urban settings to remote field sites with cellular coverage. The unobtrusive device collects data on **fine particulate matter (PM_{2.5})**, **temperature**, **relative humidity**, and **barometric pressure** and includes four low-cost gas sensors: **Ozone (O₃)**, **Nitrogen Dioxide (NO₂)**, **Sulfur Dioxide (SO₂)**, and **Carbon Monoxide (CO)**. External sensors can be added to the device for more customized designs.

A connected solar panel recharges a **LiFePO₄** battery oriented with (+) at the leaf point of the casing. Eclipse communicates via **cellular (nRF91)** and **Bluetooth** modules to transmit data to a cloud platform and data portal for your team to visualize the sensor data. The device is built for rugged outdoor operation with waterproof housing and magnetic power activation.

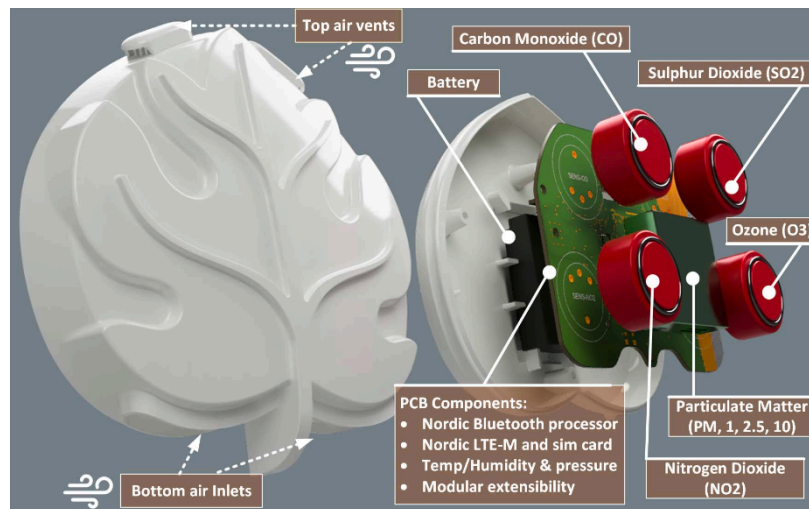


Figure Left: Components of Eclipse environmental monitoring device including air vents and air inlets allowing for cooling capabilities, bluetooth to connect to Eclipse Device Provisioning application, cellular capabilities, and software that allows for additional sensors to be added ontop of listed sensors in image.

Accessing Data

Sensor data is transmitted to the cloud via cellular data to a data visualization platform that is personalized for each community.

Offline Capabilities

The device caches records on its SPI Flash chip when there are lapses of cellular activity for 2,500 records (8 days worth of data).

Development/History

Eclipse was originally developed by Microsoft Research in 2021 for an environmental monitoring project to monitor air quality at 115 different bus stops across Chicago. After the funding ended for the Microsoft Research-specific project, the monitoring devices were revived through a second research iteration in partnership between MIT's SEED Lab and Georgia Tech's Ka Moamoa Lab.

The mission of The Eclipse Project is to offer a low-cost environmental sensor that allows communities to readily collect, organize, and visualize hyper-local environmental data to address community-led environmental questions and concerns.

Core Specifications

Category	Details
Processor	Nordic nRF91 System-on-Chip (integrated LTE-M/NB-IoT and GPS)
Connectivity	LTE-M / NB-IoT cellular, Bluetooth Low Energy (BLE)
Power Supply	Rechargeable LiFePO4 battery, solar charging capability
Charging Interface	Dedicated LiFePO4 dual-slot charger; USB jump-start support
Enclosure	Waterproof casing with silicone gasket and magnetic power switch
Firmware Management	Flashing and provisioning via Segger J-Link, nRF Command Line Tools, and Windows Provisioning App
Cloud Integration	Azure IoT Hub for device registration, telemetry, and monitoring
Diagnostics	Command Line Interface (CLI) via Provisioning App for live tracing and connectivity checks
Internal Sensors	Fine particulate matter (PM2.5), temperature, relative humidity, and barometric pressure
Four Low-Cost Gas Sensors	Carbon Monoxide (CO), Sulphur Dioxide (SO2), Ozone (O3), Particulate Matter (PM, 1, 2.5, 10), Nitrogen Dioxide (NO2)

External Capabilities	Sensor	Environmental (e.g., soil moisture, air quality, particulate matter), Hydrological (e.g., water level, flow rate), Agricultural (e.g., soil temperature, leaf wetness)
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Example Use Cases

Eclipse is designed to be a useful monitoring tool across disciplines, locations, and climates. Use cases include environmental monitoring, remote infrastructure telemetry, agricultural field sensing, smart city, and stormwater systems.

Eclipse has been used by various communities to achieve varying community-identified environmental monitoring goals. The following are examples of some of these goals:

- Illuminate the hyper-local air quality at 115 bus stops in Chicago.
- Collect hour-by-hour hyper-local air quality data to tell the story of people living near one of the largest shipping ports in the United States.